BRIEF FOR COUNTRIES & PARTNERS

WHO/ITU Reference Architecture for Digital Public Infrastructure for Health (DPI-H)

Foundational DPI → health-specific shared components → a governed, interoperable national digital health infrastructure → every programme reuses rather than rebuilds.

A joint WHO and ITU blueprint for building shared national digital health infrastructure. **DPI-H is "digital resources that are built as a shared means for fulfilling multiple health agency functions"**: interoperable, reusable, publicly governed and built on open standards. Public consultation draft V1.0; released for public comment in July 2026 (now closed) and currently undergoing revision and updates (WHO & ITU, 2026).

1 Key concepts

Shared means

Infrastructure every health actor reuses, not programme-specific apps that each rebuild the same capability.

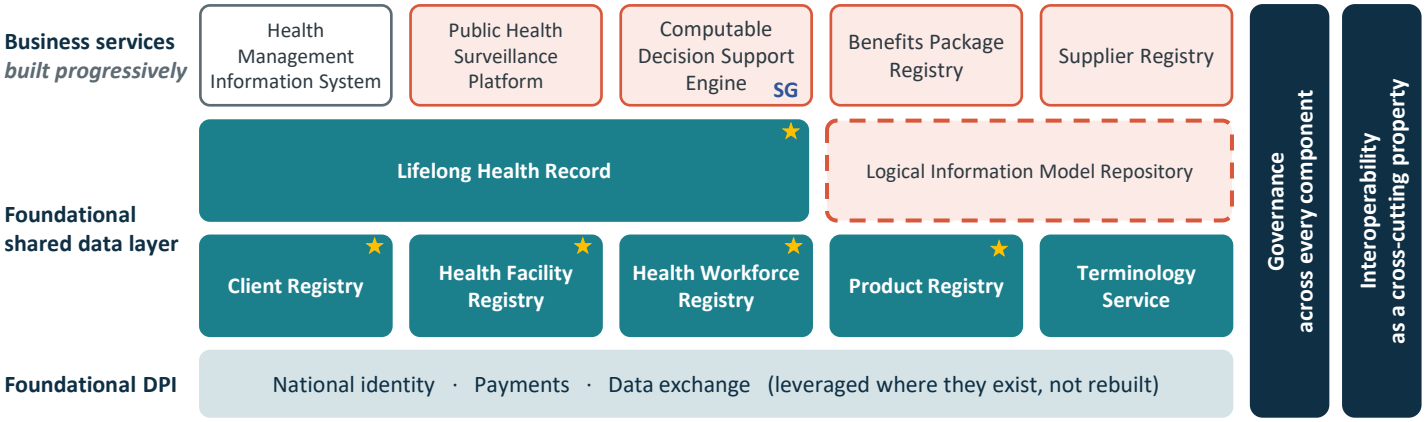
Extends foundational DPI

Builds on national identity, payments and data exchange. Adds only what health needs; rather than replicating.

Goal-driven design

Informed by global architecture frameworks. Every component traces back to priority health-sector goals.

2 The architecture at-a-glance



Teal = the DPI-H foundation that matures over time; ★ = forms **REDDHI** (Resilient Essential Data and Digital Health Infrastructure), the resilient essential foundation; **Coral** = new component introduced by DPI-H, not present in the OpenHIE architecture; **Dashed** = enabler built up as maturity and data standardisation grows. **Navy** = cross-cutting properties of the whole architecture, not single boxes to buy. **SG** = runs **WHO SMART Guidelines**: computable clinical and public-health logic, drawing on the shared DPI-H foundation.

3 Why generic DPI is not enough

Identity proves who someone is, not entitlements or consent, and can miss newborns, refugees and migrants.
→ **Client Registry**

Data exchange is a "sealed envelope": clinical meaning is not guaranteed to survive transit.
→ **Terminology and data standards**

Payments move money once a decision is made, not benefit, licensing or service checks.
→ **Benefits Package Registry**

4 Reference Architecture design principles

I	Governance	VI	Local capacity
II	Open standards	VII	Equity and inclusion
III	Interoperability	VIII	Data ownership
IV	Sustainability	IX	Trust and security
V	Modularity	X	Ethics

5 What this means for countries

Establish architecture governance and leadership

A governance body for architecture decisions, and a national digital architect (person or small team) with authority to say no.

Design data models first

Define minimum data models per registry before choosing software. A FHIR profile represents the model; it does not replace it.

Treat interoperability as a cross-cutting capability

Mix mechanisms to fit context (bus, gateways, direct, publish-subscribe), not one mandatory central purchase.

Build progressively

Start with core registries, the health record and basic terminology for priority goals. Add business services as the foundation matures.

Country signals: Rwanda embeds shared foundations in financing and accountability. Ethiopia, the Philippines and Uganda have piloted computable WHO SMART Guidelines logic in areas such as antenatal care, HIV and immunisation.

Source: World Health Organization & International Telecommunication Union. (2026). Reference architecture and guidance for digital public infrastructure for the health sector (Draft V1.0, public consultation draft).